

(ii) The slope of a $V-I$ graph gives the resistance of the conductor. The voltage shows in y -axis and current shows in x -axis.

(iii) By Ohm's law $V = IR$
 $\Rightarrow R = \frac{V}{I}$ ~ slope of $V-I$ graph

Therefore, the slope of the $V-I$ graph represents the effective resistance.

Now in series combination of the resistors all the resistance get summed up.

$$R_{eq} = R_1 + R_2 + R_3 + \dots + R_n$$

Therefore, the series combination will have the maximum resistance than the other resistors.

Hence, the line with maximum slope line C will give the series combination of the other two resistors.

12) given are,
 $u = -25 \text{ cm}$
 $f = -15 \text{ cm}$
 $h_i = 4 \text{ cm}$
 $v = ?$
 $h_o = ?$

(i) using mirror formula $= \frac{1}{v} + \frac{1}{u} = \frac{1}{f}$

$$\Rightarrow \frac{1}{v} + \frac{1}{-25} = \frac{1}{-15}$$

$$\Rightarrow \frac{1}{v} = \frac{1}{-15} + \frac{1}{25} \Rightarrow \frac{1}{v} = \frac{-5 + 3}{75} = \frac{-2}{75}$$

$$\Rightarrow \frac{1}{v} = \frac{-2}{75} \Rightarrow v = \frac{-75}{2} = -37.5$$

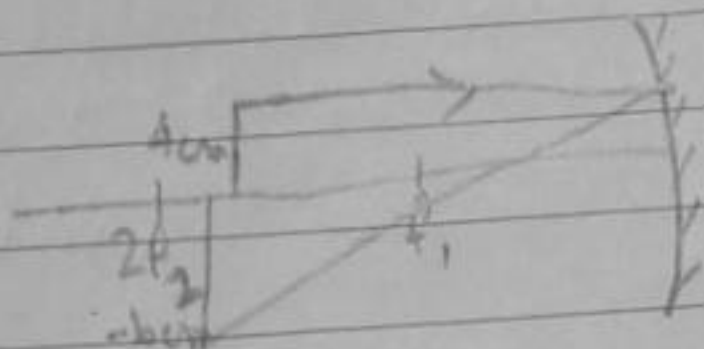
$$v = -37.5$$

(ii) $\frac{h_i}{h_o} = \frac{-v}{u} \Rightarrow \frac{4}{h_o} = \frac{37.5}{-25 \text{ cm}} \Rightarrow \frac{4}{h_o} = \frac{-100}{37.5}$

$$h_i = \frac{4 \times 37.5}{25} = 6$$

$$h_i = -\frac{150}{25} = -6 \text{ cm}$$

iii



Real, Inverted